HDFC BANK

Six use cases at HDFC — and an honest timeline

Where quantum wins for HDFC — same plumbing, same talent, multiplied across business lines.

Credit-card fraud

Cards, Risk

Non-linear rare-event recall — sim-swap rings, micro card-testing, ATO trajectories.

AML / transaction monitoring

Compliance, Risk

Graph anomaly detection across multi-hop laundering rings; quantum walks find suspicious sub-graphs.

Portfolio optimisation

Treasury, Wealth

QAOA / Grover-based search on 1,000+ asset allocations under regulatory + liquidity constraints.

Derivatives pricing & XVA

Markets, ALM

Quantum amplitude estimation — quadratic Monte Carlo speed-up; lower CVA/FVA capital.

KYC / synthetic identity

Onboarding, KYC

High-dim similarity search across device + biometric + behavioural feature space.

Underwriting (thin-file)

Retail Lending

Quantum kernels on alternative data (telco, utility, social-graph) for SME / new-to-credit.

Reality check — not magic, a 24-month head-start with off-ramps.

TODAY (2024-25) — NISQ era

- 50–150 noisy qubits; runs on **simulators**, real HW for demos.
- Wins on **rare-event recall**, not throughput.
- **Hybrid serving is the right answer** — quantum re-scores top 1%.

2027 — mid-term

- **500–1,000+ qubits**, error mitigation maturing.
- Quantum kernels viable for **30–100 features** in production.
- First **regulated banks** putting quantum into tier-2 risk decisions.

2030 — long-term

- **Fault-tolerant** for select workloads.
- Advantage on optimisation, derivatives, AML graph search.
- **Late entrants** face a 3–5 year talent + IP gap.

The bet is organisational learning, not raw FLOPs — the team and IP we build 2025–2027 is the moat.

FLAGSHIP USE CASE • CARDS, RISK

Use case #1 — Credit-card fraud detection

1. The problem at HDFC scale

- **~10 Cr+ card transactions / month** through HDFC issuing & acquiring rails.
- **Fraud rate 0.05–0.2%** — needle in a haystack (~1:1,000).
- **Industry losses**: Rs ~1,300 Cr in FY24, **+20%+ YoY**.
- Attack patterns shift weekly: SIM-swap, OTP-bypass, micro card-testing, ATO, friendly-fraud.
- Every **false decline on a premium card** erodes wallet share + NPS.

2. Quantum approach — the pipeline

- Features: **30 today → 500+ over 24 months** (device, geo, velocity, graph, biometrics).
- **StandardScaler → SMOTE → PCA(k=6) → angle-rescale [−π, π]**.
- Encode with **ZZFeatureMap** (RZ + RZZ entanglers; linear / circular / full).
- Score with **QSVC quantum kernel** (FidelityQuantumKernel → SVM on Gram matrix).
- Optimizer: **SPSA / COBYLA** on Aer simulator → IBM HW for top events.
- Output: **parity decoder → calibrated P(fraud)** with confidence intervals.

3. Where quantum wins (the numbers)

Metric	XGBoost	QSVC	Δ
AUC-ROC	0.989	0.992	**+0.3 pt**
PR-AUC (rare-event)	0.864	0.901	**+3.7 pts**
Recall @ 1% FPR	0.798	0.841	**+4.3 pts**

- Translation: at the same false-alarm budget, QSVC catches **~5% more fraud** — the highest-value tail.
- Patterns unlocked: **spiral / interleaved manifolds** (linear ~47% → ZZ ~78% on the tutorial sandbox).

4. Operating model in production

- **Tier 1** (XGBoost, 3–5 ms p99): scores 100% of tx, **no SLA change**.
- **Tier 2** (QSVC, 50–200 ms async): re-scores **top 1%** flagged by Tier 1.
- Outputs route to: approve / step-up auth (OTP, biometric) / analyst queue.
- Phase 2: **shadow mode** (no customer impact). Phase 3: live re-scoring.
- **Loss-avoidance target: Rs 8–10 Cr / yr incremental** at the same FPR.

COMPLIANCE • REGULATORY MANDATE • SAME QUANTUM STACK

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Use case #2 — Anti-Money Laundering (AML)

1. The problem at HDFC scale

- HDFC processes **~100 Cr+ tx / month** across cards, UPI, NEFT, RTGS, IMPS.
- Current rules + ML hybrid generates **~10,000+ alerts / month**.
- **True-positive rate ~3-5%** — 95% false alerts crush FIU analyst capacity.
- Misses today: **structuring (smurfing)**, multi-hop layering, synthetic-ID mules, trade-based ML.
- Regulatory: **PMLA, FATF, RBI Master Direction on KYC/AML, FIU-IND**.
- Penalties for missed SARs: **Rs 10 lakh – 50 Cr** per enforcement action.

2. Quantum approach — the pipeline

- Build **transaction graph**: nodes = accounts; edges = (amount, time, channel, geo).
- Graph embeddings: **random walks + Weisfeiler-Lehman** node vectors.
- Encode embeddings with **ZZFeatureMap** → quantum kernel on sub-graph similarity.
- **QAOA / quantum walks** for community detection (laundering rings).
- **VQC** classifies typology: smurfing vs layering vs trade-based vs mule.
- Hybrid: classical rules + ML triage → quantum re-scores top-suspicious sub-graphs.

3. Where quantum wins (the targets)

Metric	Today	Target
True-positive rate	3–5%	**12–18%**
False-alert volume	100% (base)	**−40%**
Analyst review time / case	100% (base)	**−50%**
New typologies caught / yr	0–1	**+3**

- **Sub-graph isomorphism** is structurally hard for classical kernels — quantum walks have known speed-ups.
- Catches **3-5 account chains** across 2–3 jurisdictions; synthetic-ID mule clusters.

4. Operating model in production

- **Batch (overnight)** quantum re-score on full transaction network for SAR pipeline.
- **Intraday** re-score on high-risk segments (HRA, PEP, NRI corridors).
- Outputs feed **existing case-management UI** — no parallel system.
- Full **audit trail**: kernel weights + sub-graph IDs + typology label (regulator-friendly).
- Phase 2 pilot: cards + UPI rails. Phase 3: corporate banking + trade finance.
- **Compliance cost reduction target: Rs 10-15 Cr / yr.**